

Karan Shah — Curriculum Vitæ

CONTACT INFORMATION

Mail: shah_karan@outlook.com *Phone:* +14044650213
Web: <https://www.karan.sh> || *GitHub:* [karanprime](#) || *Twitter:* [@ReKarantNetwork](#)

EDUCATION

PhD Candidate, Computer Science *Aug 2021 - Aug 2026*
Center for Advanced Systems Understanding (CASUS), Görlitz, Germany
Technische Universität Dresden, Dresden, Germany

MS Computational Science & Engineering *Dec 2020*
Primary Focus: Machine Learning applied to data-intensive *Physics* problems
BS Computer Science (Threads: Intelligence, Modeling-Simulation) *May 2018*
BS Physics
Thesis: *“Analysis of Uncertainty Quantification of Machine Learned Density Functionals”*
Georgia Institute of Technology, Atlanta, GA USA

EXPERIENCE

Roblox Corporation, San Mateo, CA USA
Manager: Dr. Nick Burgess
Physics - Simulation Solver Team, Engine Group
Senior Physics Programmer *Nov 2025 - Present*
Research Intern *Sep 2024 - Dec 2024*
Project: Developing ML techniques for rigid body simulations in the game engine. Using generative ML models for reduced-order modeling to accelerate 3D game engine performance.

Center for Advanced Systems Understanding (CASUS), Görlitz, Germany
Helmholtz-Zentrum Dresden-Rossendorf e.V. (HZDR)
Supervisor: Dr. Attila Cangi
Machine Learning for Materials Design Department
Doctoral Researcher *Aug 2021 - Oct 2025*
Project: A simulation framework for quantum dynamics based on physics informed neural networks. Subprojects include ML accelerated PDE solvers, synthetic ML generated data to accelerate surrogate model training. Funded by Helmholtz AI.

Lawrence Livermore National Laboratory, Livermore, CA USA
Hosted by: Dr. Michael Schneider
Astronomy and Astrophysics Analytics Group
Graduate Intern, Physics Division *May 2019 - Aug 2019*
Technical Scholar, Physics Division *Aug 2017 - May 2019*
Intern, Data Science Summer Institute *May 2017 - Aug 2017*
Projects: 1) Gaussian Processes with neural network equivalent kernels to estimate cosmological parameters from mass density fields with uncertainty quantification
2) Probabilistic Inference of Cosmic Shear & Intrinsic Galaxy Properties through Hierarchical Graphical Models. Used MCMC techniques to determine cosmic shear and galaxy morphology (for LSST)

Georgia Institute of Technology, Atlanta, GA USA

Medford Group, School of Chemical & Biomolecular Engineering *Jan 2017 - Aug 2019*
Advisor: Dr. Andrew Medford
Project: Determination of Exchange Correlation Functionals through Deep Learning
Using ensembles of neural networks to build surrogate density functionals

PUBLICATIONS

* DENOTES EQUAL CONTRIBUTION

- **Shah, K.**, Cangi, A., *Machine Learning Time Propagators for Time-Dependent Density Functional Theory Simulations*, Machine Learning: Science and Technology, 2026. DOI: 10.1088/2632-2153/ae68d4. (2 citations)
- Cangi, A., Fiedler, L., Brzoza, B., **Shah, K.**, et al., *Materials Learning Algorithms (MALA): Scalable machine learning for electronic structure calculations in large-scale atomistic simulations*, Computer Physics Communications, Volume 314, 2025, 109654, ISSN 0010-4655. (9 citations)
- **Shah, K.**, Cangi A., *Accelerating Electron Dynamics Simulations through Machine Learned Time Propagators*, ICML 2024 AI for Science Workshop. (3 citations)
- **Shah, K.**, Butler, J., Knaub, A., Ratcliff, W., Zenginoglu, A., Soltanieh-ha, M., *Data Science Education in Undergraduate Physics: Lessons Learned from a Community of Practice*, Am. J. Phys. 92(9), 655–662, 2024. (4 citations)
- Martinetto, V.* , **Shah, K.***, Cangi, A., Pribram-Jones, A., *Inverting the Kohn-Sham equations with physics-informed machine learning*, MLST 5(1), 015050, 2024. (10 citations)
- **Shah, K.**, Stiller, P., Hoffmann, N., Cangi A., *Physics-Informed Neural Networks as Solvers for the Time-Dependent Schrödinger Equation*, NeurIPS 2022 ML4PS Workshop. (23 citations)
- Fiedler, L., **Shah, K.**, Bussmann, M., Cangi A., *Deep dive into machine learning density functional theory for materials science and chemistry*, PRMaterials 6(4), 040301, 2022. (160 citations)
- Dzanic, T., **Shah, K.**, Witherden, F., *Fourier Spectrum Discrepancies in Deep Network Generated Images*, NeurIPS 2020, Vol. 33, pp. 3022–3032. (250 citations)

*

BOOK CHAPTERS

- Fiedler, L., **Shah, K.**, & Cangi A., *Machine-Learning for Static and Dynamic Electronic Structure Theory*, in *Machine Learning in Molecular Sciences*, Springer, 2023. (3 citations)

PEER REVIEW ACTIVITIES (INVITED)

• Conferences & Workshops:

– Conferences

- * AAAI 2026: Reviewer (4 papers)
- * ACM Conference on AI and Agentic Systems: (Demonstration Track) Reviewer: (4 demos/papers)
- * NeurIPS 2026: (Main Conference): Reviewer (4 papers; review period ongoing)
- * NeurIPS 2025: (Main Conference; Benchmarks & Datasets): Reviewer (4 papers)
- * ICML 2025: Reviewer (3 papers; 4 invited, 1 withdrawn)
- * ICLR 2025: Reviewer (1 paper)
- * NeurIPS 2024: Reviewer (1 paper)

– Workshops

- * ICML 2026 AI4Science Workshop: Reviewer (3 papers)
- * ICLR 2026 GRaM Workshop: Reviewer (2 papers)
- * ICLR 2026 DeLTa Workshop: Reviewer (4 papers)
- * CVPR 2026 SynData4C Workshop: Reviewer (5 papers)
- * NeurIPS 2025 ML4PS Workshop: Reviewer (4 papers)
- * ICLR 2025 Workshop on Deep Generative Models in Machine Learning: Theory, Principle and Efficacy: Reviewer (3 papers)
- * NeurIPS 2023 ML4PS Workshop: Reviewer (2 papers)
- * NeurIPS 2022 ML4PS Workshop: Reviewer (2 papers)
- * NeurIPS 2022 SyntheticData4ML Workshop: Reviewer (1 paper)
- * ML Reproducibility Challenge 2021: Reviewer (2 papers); Outstanding Reviewer Award

• Journals:

- Journal of Open Source Software: 1 paper (2026)
- American Journal of Physics: 1 paper (2025)
- Machine Learning: Science and Technology (MLST): 1 paper (2025)
- RSC Digital Discovery: 1 paper (2025)
- Physiological Measurement (IOP; survey on physics-informed ML): 1 paper (2025); IOP Outstanding Reviewer Award 2025
- New Journal of Physics: 1 paper (2024); IOP Trusted Reviewer 2024

HONORS AND AWARDS

- Travel Grant, NHR (German National HPC Alliance) Conference 2023, September 2023
- Elected Member, American Physical Society-Group on Data Science (APS-GDS) [Executive Committee](#), June 2023 - March 2025
- APS [Data Science Education & Community of Practice Fellowship 2022-2023, 2023-2024](#)
- IOP Publishing Outstanding Reviewer Award, 2025
- Outstanding Reviewer Award, ML Reproducibility Challenge 2021
- Datmo Applied Machine Learning Fellowship, December 2017
- Amazon Web Services Research Grant, September 2017 (GT Data-Driven Education team)
- President's Undergraduate Research Award: Fall 2017, Fall 2016
- Fellow, Data Science Summer Institute, LLNL, Summer 2017
- Student Travel Awards: JupyterCon 2017 (NYC), WSSSPE 2016 (Manchester, UK)
- Top 10 percentile in Indian National Astronomy Olympiad, 2012

POSTERS AND PRESENTATIONS

- Lecture, Psi-k Conference, Lausanne, Switzerland, August 2025.
- Poster, Roblox Research Showcase, San Mateo, USA, October 2024.
- Poster, ICML 2024 AI for Science Workshop, Vienna, Austria, July 2024.
- Lecture, DPG Spring Meeting, Berlin, Germany, March 2024.
- Invited Tutorial, APS March Meeting, Minneapolis, MN, USA, March 2024.
- Lecture, APS March Meeting, Minneapolis, MN, USA, March 2024.
- Lecture, NHR Conference, Berlin, Germany, September 2023.
- Poster, TDDFT School and Workshop, Newark, USA, July 2023.
- Invited Lecture, DSECOP Workshop, College Park, MD, USA, June 2023.
- Poster, DPG Spring Meeting, Dresden, Germany, March 2023.
- Poster, NeurIPS 2022 Machine Learning for Physical Sciences Workshop, New Orleans, USA, December 2022.
- Invited lecture, Research Seminar: Many-Body Theory, First principle Simulationen und Machine Learning, Kiel University, Rostock, Germany, December 2022.
- Poster, HZDR Doctoral Seminar, Wrocław, Poland, October 2022.
- Poster, Big Data Analytical Methods for Complex Systems, Wrocław, Poland, October 2022.
- Lecture, Multiscale Modeling of Matter under Extreme Conditions, Görlitz, Germany, September 2022.
- Poster, Strongly Coupled Coulomb Systems, Görlitz, Germany, July 2022.
- Poster, Helmholtz AI Conference, Dresden, Germany, June 2022.
- Invited Lecture, DSECOP Workshop, College Park, MD, USA, June 2022.

- Lecture, APS March Meeting, Chicago, USA, March 2022.
- Poster, DFT Methods for Matter under Extreme Conditions, Görlitz, Germany, February 2022.

TEACHING
EXPERIENCE

Fellow, Data Science Education Community of Practice *Feb 2022 - Dec 2024*
Created multiple open-source pedagogical modules for integrating machine learning topics into the undergraduate physics curriculum, as part of a competitive APS fellowship. Link: [GitHub](#)

Graduate Teaching Assistant, College of Computing, Georgia Tech *Aug 2018 - May 2020*
TA for Junior Level *CS 3510 - Design-Analysis of Algorithms* *S'20*
TA for Graduate Level *CSE 6730 - Modeling & Simulation* *S'19*
TA for Senior Level *CS 4510 - Automata & Complexity* *F'18*

SERVICE AND
LEADERSHIP

Early Career Member-at-Large, APS GDS Executive Committee *June 2023 - March 2025*
Doctoral Representative, HZDR, Dresden, Germany *Feb 2022 - Present*
Member of Taskforce Promovierende (Graduate Student Education), HZDR, Dresden, Germany *Apr 2022 - Dec 2023*
Volunteer, ICML 2020, Remote *July 2020*
Volunteer, ICLR 2019, New Orleans, LA *May 2019*
Senator, Graduate Student Senate, Georgia Tech *Sept 2018 - May 2019*
Representing Computational Science & Engineering in the Student Government Association.
Reviewer, President's Undergraduate Research Award (PURA) *May 2018 - Dec 2020*
Reviewed Physics and CS research proposals for PURA, a competitive undergraduate research award.